

COMPOSITE MATERIAL DATASHEETS CFRP G0904/M20-PW

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	G0904/M20-PW
Commercial name	Hexcel HexPly® M20/40%/G904
Specifications	Locals: Airbus: AIMS08-03-001 Others:
Description	Carbon Fiber Reinforced Epoxy Prepreg for Repair 120°C Curing class / Plain Weave Fabric, 193 g/m ²

REFERENCES

- QTRSM0299 Is. 6 'Qualification Test Report of G0904/M20 Plain Weave Fabric'

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.211 mm	Coefficients of Thermal Expansion (1/°C)	Range	-55..20°C	20..70°C	70..120°C
Density	1530 kg/m ³		α_1	-	-	-
Tg-onset (wet)	123 °C		α_2	-	-	-
Max. Operational Limit (MOL)	90 °C		α_3	-	-	-

ELASTIC MODULI

Property	Units	Design Values (MPa) per condition						Buckling K _B
		RT/Dry	55°C/W	80°C/W	90°C/W	120°C/W	AII (*)	
E ₁	MPa	58600	58600	58600	-	58600	58600	0.835
E ₂	MPa	58600	58600	58600	-	58600	58600	
G ₁₂	MPa	4700	4200	3300	-	2100	3300	
V ₁₂	-	0.05						-

(*) Consolidated design values applicable on any condition below MOL, used to normalize strengths into allowable strains

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa) per condition				B-basis K _B	E-KDF		
	RT/Dry	55°C/W	80°C/W	120°C/W		55°C/W	80°C/W	120°C/W
F _{1t}	785.8	766.0	754.0	688.0	0.916	0.975	0.960	0.876
F _{1c}	785.6	571.0	484.5	274.5	0.912	0.727	0.617	0.349
F _{2t}	871.0	-	-	-	0.916	-	-	-
F _{2c}	827.0	-	475.0	-	0.912	-	0.574	-
F ₁₂	116.2	62.0	56.2	45.0	0.874	0.534	0.484	0.387

Property	Mean Values (MPa) per condition				B-basis K _B	E-KDF		
	RT/Dry	55°C/W	80°C/W	120°C/W		55°C/W	80°C/W	120°C/W
F ₁₃	77.1	-	42.1	-	0.878	-	0.546	-

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DEFENCE AND SPACE

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	B-basis Values per condition			
	RT/Dry	55°C/W	80°C/W	120°C/W
ϵ_{tall} ($\mu\epsilon$)	12400	12000	11800	-
ϵ_{call} ($\mu\epsilon$)	9400	8700	7400	-

Remarks:

- Tension values corrected with QI and directed laminates data (not applicable for directions with less than a 25% of the plies aligned)
- Compression values right from UD data

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			55°C/W	80°C/W	120°C/W
OHT_{QI} (MPa)	318.7	0.919	-	1.000	-
OHC_{QI} (MPa)	280.6	0.859	0.859	0.730	0.533
FHT_{QI} (MPa)	334.0	0.901	-	-	-
FHC_{QI} (MPa)	378.0	0.848	-	0.807	-
F_{BR-QI} (MPa)	952.0	0.859	-	0.829	0.631

COMPOSITE DAMAGE TOLERANCE

To be completed in further revisions.